Supplemental Material for 04/15 Book Reading Seminar

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Abstract

This is the supplemental material for the book reading seminar.
Please refer to the textbook and the whiteboard for the main content.
If necessary, please also refer to my repository.

Our goal

$$\min_{x \in \mathcal{M}} f(x) \tag{3.1}$$

Unit sphere

$$\mathrm{S}^{d-1} := \{x \in \mathbb{R}^d : x^\top x = 1\} \tag{3.2}$$

Personal Opinion: The sphere is actually a quite good example of a manifold, but I recommend NOT to rely too much for intuition on the sphere.

Smoothness

$$f \text{ is smooth} \iff f \text{ is of class } C^\infty.$$

Adapted from Wikipedia:

$$\begin{gathered}(U \subseteq \mathbb{R}^d \text{ is open}, \quad f\colon U \to \mathbb{R}, \quad k \in \mathbb{N}) \\ f \text{ is of class } C^k \text{ on } U \\ \iff \\ \forall \alpha \in \mathbb{N}^d, |\alpha| \leq k, \quad \forall y \in U, \quad \frac{\partial^\alpha f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}(y_1, y_2, \ldots, y_d) \text{ exists and is continuous.}\end{gathered}$$

It is known that the smoothness is also equivalent to the existence and continuity of the k-th Fréchet derivatives:

$$\begin{gathered}(U \subseteq \mathbb{R}^d \text{ is open}, \quad f\colon U \to \mathbb{R}, \quad k \in \mathbb{N}) \\ f \text{ is of class } C^k \text{ on } U \\ \iff\end{gathered}$$

$$\forall j \in \{1, 2, \ldots, k\}, \quad \forall y \in U, \quad \exists A \colon U \to \mathbb{R}^d, \quad \lim_{\|h\| \to 0} \frac{\|f(y+h) - f(y) - A(h)\|}{\|h\|} = 0.$$

Recall: In general, partial differentiability does not imply total differentiability (Fréchet differentiability). However, the above statement holds. Why? This is because the smoothness requires not only existence but also continuity of the partial derivatives. The continuity of the partial derivatives ensures that the function behaves well enough to guarantee total differentiability.

Wiki: Smoothness Wiki: Fréchet derivative

Def 3.10

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space}, \quad \emptyset \neq \mathcal{M} \subseteq \mathcal{E})$$
$$\mathcal{M} \text{ is a smooth } \textit{embedded submanifold} \text{ of } \mathcal{E} \text{ of dimension } n$$
$$\iff$$
$$\Big(n = d \text{ and } \mathcal{M} \text{ is open in } \mathcal{E} \text{ (This } \mathcal{M} \text{ is } \textit{open submanifold}\text{)}\Big) \text{ or}$$
$$\Big(n = d - k \ (k \geq 1) \text{ and } \forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^k \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_k \iff y \in \mathcal{M}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = k)\Big).$$

Example: S^2 in $\mathbb{R}^3$

$$\mathrm{S}^2 = \{x \in \mathbb{R}^3 : x^\top x = 1\} = \{x \in \mathbb{R}^3 : h(x) = 0\}, \quad h(x) = x^\top x - 1.$$

We can check the conditions in Def 3.10:

$$n = 2, \quad k = 1, \quad d = 3,$$
$$h \colon U_x = \mathbb{R}^3 \subseteq \mathcal{E} = \mathbb{R}^3 \to \mathbb{R}, \quad h(x) = x^\top x - 1 \text{ is smooth},$$
$$h(y) = 0 \iff y^\top y = 1 \iff y \in S^2,$$
$$\operatorname{rank}(\mathrm{D}h(x)) = \operatorname{rank}(2x^\top) = 1 \quad (\text{since } x \neq 0).$$

Thus, S^2 is a smooth embedded submanifold of $\mathbb{R}^3$ of dimension $n = 2$.

Def 3.11 The textbook's definition is somehow simplified than the general one:

$$(\mathcal{E}_1, \mathcal{E}_2 \text{ are linear spaces}, \quad U \subseteq \mathcal{E}_1, V \subseteq \mathcal{E}_2 \text{ are open})$$
$$F \colon U \to V \text{ is a diffeomorphism}$$
$$\iff$$
$$F \text{ is a bijection and } F \text{ and } F^{-1} \text{ are smooth.}$$

Wiki: Diffeomorphism

Theorem 3.12

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space})$$
$$\mathcal{M} \subseteq \mathcal{E} \text{ is a } n = d - k\text{-dimensional smooth embedded submanifold of } \mathcal{E}$$
$$\Longleftrightarrow$$
$$\forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open},$$
$$\exists F \colon U_x \to V \text{ is a diffeomorphism},$$
$$F(\mathcal{M} \cap U_x) = (\mathbb{R}^n \otimes 0_k) \cap V.$$

Rewrite of Theorem 3.12

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space})$$
$$\mathcal{M} \subseteq \mathcal{E} \text{ is a } n = d - k\text{-dimensional smooth embedded submanifold of } \mathcal{E}$$
$$\Longleftrightarrow$$
$$\forall x \in \mathcal{M}, \quad \exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open},$$
$$\exists F^{-1} \colon V \to U_x \text{ is a diffeomorphism},$$
$$F^{-1}((\mathbb{R}^n \otimes 0_k) \cap V) = \mathcal{M} \cap U_x.$$

Non-degenerate $\mathrm{D}F^{-1}$

$$(\mathrm{D}F(x))^{-1} = DF^{-1}(F(x)) \tag{3.20}$$

Since $\mathrm{D}F^{-1}(F(x)) \cdot \mathrm{D}F(x) = \mathrm{D}(F^{-1} \circ F)(x) = \mathrm{D}(\mathrm{id})(x) = I_d$, this is non-degenerate.

Def 3.14

$$(\mathcal{E} \text{ is a linear space}, \quad \mathcal{M} \subseteq \mathcal{E}, \quad x \in \mathcal{M})$$
$$\mathrm{T}_x\mathcal{M} \coloneqq \{c'(0) \mid \exists \varepsilon > 0, \quad \exists c \colon (-\varepsilon, \varepsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x\} \tag{3.23}$$

Tangent space is a subset of the linear space

$$\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}$$

This is obvious but not so trivial.

$$c'(0) = \lim_{t \to 0} \frac{c(t) - c(0)}{t}$$

Since $c(t) \in \mathcal{M} \subseteq \mathcal{E}$ and $c(0) = x \in \mathcal{M} \subseteq \mathcal{E}$, we have $c(t) - c(0) \in \mathcal{E}$ for all t. Therefore, the limit is also in $\mathcal{E}$. Hence, $\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}$.
Question: $c'(0) \in \mathcal{M}$? The answer is no. $\mathcal{M}$ is not closed under addition and scalar multiplication.

Theorem 3.15

$$(\mathcal{E} \text{ is a } d\text{-dimensional linear space}, \quad \mathcal{M} \subseteq \mathcal{E} \text{ is a smooth embedded submanifold of dimension } n, \quad x \in \mathcal{M})$$

$$\begin{cases} n = d & \implies \mathrm{T}_x\mathcal{M} = \mathcal{E} \\ n = d - k \ (k \geq 1) & \implies \mathrm{T}_x\mathcal{M} = \mathrm{Ker}(\mathrm{D}h_x(x)) \quad (\text{where } h_x \text{ is in Def 3.10}) \end{cases}$$

$\mathrm{T}_x\mathcal{M} \subseteq \mathcal{E}$ is already stated. We will show $\mathrm{T}_x\mathcal{M} \supseteq \mathcal{E}$.

For the case $\mathcal{M}$ is a open submanifold: It is suffices to show that

$$\forall v \in \mathcal{E}, \quad \exists \epsilon > 0, \quad \exists c\colon (-\epsilon, \epsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x, \quad c'(0) = v.$$

Since $\mathcal{M}$ is open and $x \in \mathcal{M}$, we have

$$\exists \epsilon > 0, \quad \{x + tv : t \in (-\epsilon, \epsilon)\} \subseteq \mathcal{M}.$$

Then, $c(t) := x + tv$ satisfies the required conditions.

For the case $n = d - k$:
Step1: $\mathrm{T}_x\mathcal{M} \subseteq \mathrm{Ker}(\mathrm{D}h_x(x))$. Recall the Def 3.10.

$$\forall v \in \mathrm{T}_x\mathcal{M}, \quad \exists \epsilon > 0, \quad \exists c\colon (-\epsilon, \epsilon) \to \mathcal{M} \text{ is smooth}, \quad c(0) = x, \quad c'(0) = v.$$
$$\forall c(t) \in U_x, \quad c(t) \in \mathcal{M} \implies h_x(c(t)) = 0_k$$
$$0_k = \frac{\mathrm{d}}{\mathrm{d}t} h_x(c(t)) = \mathrm{D}h_x(c(t))[c'(t)] \implies \mathrm{D}h_x(x)[v] = 0 \implies v \in \mathrm{Ker}(\mathrm{D}h_x(x)).$$

Step2: $\mathrm{T}_x\mathcal{M} \supseteq$ (n-dimensional subspace of $\mathcal{E}$). Recall the Theorem 3.12.

$$\exists U_x \subseteq \mathcal{E} \text{ is open}, \quad \exists V \subseteq \mathbb{R}^d \text{ is open},$$
$$\exists F^{-1}\colon V \to U_x \text{ is a diffeomorphism},$$
$$F^{-1}((\mathbb{R}^n \otimes 0_k) \cap V) = \mathcal{M} \cap U_x.$$

We can construct a curve c to use the definition of $\mathrm{T}_x\mathcal{M}$ as follows:

$$\forall u \in \mathbb{R}^n, \quad \exists \epsilon > 0, \quad c\colon (-\epsilon, \epsilon) \to \mathbb{R}^d, \quad c(t) = F^{-1}(F(x) + t(u \otimes 0_k)) \text{ is smooth},$$
$$c(0) = F^{-1}(F(x)) = x, \quad c'(0) = \mathrm{D}F^{-1}(F(x))[u \otimes 0_k].$$

Thus, $\mathrm{T}_x\mathcal{M}$ contains the n-dimensional subspace, which is $\mathrm{Ker}(\mathrm{D}h_x(x))$.

Example 3.19

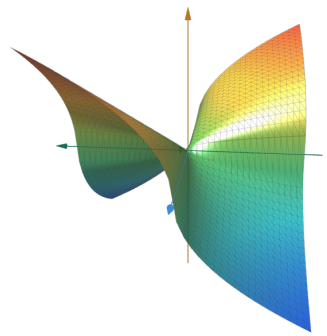

Figure 1: Visualization of $x_1 x_3 = x_2^2$ in $\mathbb{R}^3$. The orange axis represents x_2.

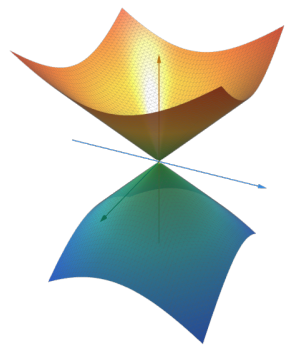

Figure 2: Visualization of $z_1^2 + z_2^2 = z_3^2$ in $\mathbb{R}^3$. The orange axis represents z_3.

We show the visualization in figs. 1 and 2.

Proposition 3.20 and Exercise 3.25

$$\Big(\mathcal{E}, \mathcal{E}' \text{ are } d, d'\text{-dimensional linear spaces,}$$
$$\mathcal{M} \subseteq \mathcal{E}, \mathcal{M}' \subseteq \mathcal{E}' \text{ are smooth embedded submanifolds of dimension } n, n'\Big)$$
$$\begin{cases} \mathcal{M} \times \mathcal{M}' \text{ is a smooth embedded submanifold of } \mathcal{E} \times \mathcal{E}' \\ \mathrm{T}_{(x,x')}(\mathcal{M} \times \mathcal{M}') = \mathrm{T}_x\mathcal{M} \times \mathrm{T}_{x'}\mathcal{M}' \end{cases}$$

First part: $n = d$ and $n' = d'$ case: $\mathcal{M} \times \mathcal{M}'$ is open in $\mathcal{E} \times \mathcal{E}'$ by the definition of the product topology.
$n = d$ and $n' = d' - k'$ case: omitted.
$n = d - k$ and $n' = d' - k'$ case:

$$\forall (x, x') \in \mathcal{M} \times \mathcal{M}', \quad \exists U_{x,x'} := U_x \times U_{x'} \subseteq \mathcal{E} \times \mathcal{E}' \text{ is open,}$$

$$\exists h_{x,x'} \colon U_{x,x'} \to \mathbb{R}^{k+k'} \text{ is smooth defined by } h_{x,x'}(y,y') = (h_x(y), h_{x'}(y')),$$
$$\left(\forall (y,y') \in U_{x,x'}, \quad h_{x,x'}(y,y') = 0_{k+k'} \iff (y,y') \in \mathcal{M} \times \mathcal{M}' \text{ and}\right.$$
$$\left.\left(\operatorname{rank}(\mathrm{D}h_{x,x'}(x,x')) = \operatorname{rank}\left(\begin{pmatrix} \mathrm{D}h_x(x) & 0 \\ 0 & \mathrm{D}h_{x'}(x') \end{pmatrix}\right) = k + k'\right)\right).$$

Second part: Similar. Omitted.

Proposition 3.23 and Example 3.26 By definition. Omitted.

Example 3.27

$$\mathcal{X} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^2\}$$

We can check the conditions in Def 3.10. Note that $\mathcal{E} = \mathbb{R}^2$ with $d = 2$.

$n = d$ **case:** Since $\mathcal{X}$ is not open in $\mathbb{R}^2$, $\mathcal{X}$ is not an open submanifold.

$n = d - k$ $(k \geq 0)$ **case:** Assume that

$$\forall x \in \mathcal{X}, \quad \exists U_x \subseteq \mathbb{R}^2 \text{ is open}, \quad \exists h_x \colon U_x \to \mathbb{R}^k \text{ is smooth},$$
$$(\forall y \in U_x, \quad h_x(y) = 0_k \iff y \in \mathcal{X}) \text{ and } (\operatorname{rank}(\mathrm{D}h_x(x)) = k).$$

Consider lines $x_1 = x_2$ and $x_1 = -x_2$. We have $h_x(y) = 0$ for any point y on these lines. Thus, $\mathrm{D}h_x(x)\begin{bmatrix}1\\1\end{bmatrix} = 0$ and $\mathrm{D}h_x(x)\begin{bmatrix}1\\-1\end{bmatrix} = 0$. This means that $\mathrm{D}h_x(x)$ is degenerate. However, $\operatorname{rank}(\mathrm{D}h_x(x)) = k \geq 1$, which is a contradiction. Thus, $\mathcal{X}$ is not a smooth embedded submanifold of $\mathbb{R}^2$.

Example 3.28

$$\mathcal{C} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^3\}$$

We can check the conditions in Def 3.10. Note that $\mathcal{E} = \mathbb{R}^2$ with $d = 2$.

$n = d$ **case:** Since $\mathcal{C}$ is not open in $\mathbb{R}^2$, $\mathcal{C}$ is not an open submanifold.

$n = d - k$ $(k \geq 0)$ **case:** Let us compute the tangent space at the origin $0 \in \mathcal{C}$:

$$\mathrm{T}_0\mathcal{C} = \{c'(0) \mid \exists \varepsilon > 0, \quad \exists c \colon (-\varepsilon, \varepsilon) \to \mathcal{C} \text{ is smooth}, \quad c(0) = 0\}.$$

On $\mathcal{C}$, we can only approach the origin along the positive x_2-axis. Thus, $c'(0) = 0$, otherwise, c cannot be smooth. (we can show it by contradiction.) Thus, $\mathrm{T}_0\mathcal{C} = \{0\}$, which is a degenerate subspace of $\mathbb{R}^2$. However, $\operatorname{rank}(\mathrm{D}h_0(0)) = k \geq 1$, which is a contradiction. Thus, $\mathcal{C}$ is not a smooth embedded submanifold of $\mathbb{R}^2$.

Example 3.29

$$\mathcal{P} := \{x \in \mathbb{R}^2 : x_1^2 = x_2^4\}$$

We can check the conditions in Def 3.10. Note that $\mathcal{E} = \mathbb{R}^2$ with $d = 2$.